KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) As an application developer, why do you choose to run an application over UDP rather than TCP. (06) [CO1]
   b) Explain the concept of piggybacking in data communication. Discuss how it affects network congestion. (05) [CO2]
   c) Discuss how TCP detects congestion in a network. Describe the congestion control techniques used in TCP to manage and avoid congestion. (12) [CO3]
   d) An organization's network is connected to the internet through a 25 Mbps access link. Users request objects of average size 800 Kbits at 30 requests/second, with average internet delay 1.5 seconds. The organization's internal LAN has a speed of 1 Gbps. To reduce response time, the organization considers: (12) [CO4]
      Option 1: Upgrade the access link to 100 Mbps.
      Option 2: Install a local web cache that serves 40% of requests.
      Compare the average end-to-end delay for both options and recommend which provides better performance and is more cost effective.

2. a) Explain how a TCP connection is established, detailing the role of SYN and ACK flags. Justify why a two-way handshake is not sufficient for reliable communication. (08) [CO2]
   b) Describe why flow control is required in TCP implements flow control between sender and receiver, specifying the conditions that must be met, and illustrate your answer with a suitable example. Additionally, explain the concept of silly window syndrome. (13) [CO3]
   c) In TCP, *EstimatedRTT*, *DevRTT* and *TimeoutInterval* adapt to changing network conditions. Suppose, three successive *SampleRTT* measurements are recorded as 110 ms, 95 ms, and 120 ms. Given: initial $EstimatedRTT = 100\ ms$, initial *DevRTT* = 5 ms, $\alpha = 0.125$, and $\beta = 0.25$ (14) [CO3]
      i) Compute *EstimatedRTT*, *DevRTT* and *TimeoutInterval* after each *SampleRTT*
      ii) Explain why it is not advisable to use the RTT of retransmitted packets when updating *TimeoutInterval*

3. a) Discuss why DNS is used in networking. Illustrate the hierarchy of DNS servers and describe how a DNS query is resolved. (09) [CO2]
   b) Explain the working of the Selective Repeat (SR) protocol. Given a sequence number space of 4 and a sender window size of 3, determine whether the window size is appropriate for SR protocol. Discuss the problems that can occur if the window size exceeds the allowed limit with a timeline diagram. (13) [CO3]
   c) Are IP address alone sufficient for communication between applications? Suppose, you have a web server that hosts three web applications on ports 3000, 5000, and 8000, but for some reasons users can't access them via standard HTTP/HTTPS requests using the domain name. Identify the underlying issue and recommend an appropriate solution to enable access via standards ports. Illustrate your approach with a request-routing diagram. (13) [CO4]

4. a) While searching for new pair of shoes, you visited multiple e-commerce sites and may or may not have completed a purchase. Later, when browsing a social media platform, you noticed advertisements for the same types of shoes you had viewed earlier. Explain the underlying mechanism that enables such personalized advertisements across different websites. (10) [CO3]
   b) Explain the rules for generating acknowledgements in TCP and discuss the fast retransmit mechanism. Illustrate your answer with suitable examples. (14) [CO3]
   c) Describe how man-in-the-middle attack hinders the authentication process (ap5.0) that uses a nonce and public key cryptography. (11) [CO3]



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## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the necessity of a trunk port in VLAN. Compare "Router-on-a-Stick" and "Layer 3 Switch (SVI)" based on following criteria: (10) [CO1]
   i) Scalability
   ii) Performance/Latency
   iii) Implementation Complexity
   iv) Cost

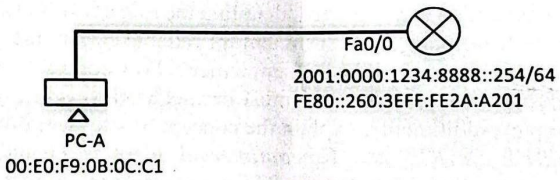
   b) Explain HOL Blocking. Analyze the major switching techniques with their benefits and drawbacks. (10) [CO1]

   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14 bytes header and has an MTU of 1024 bytes; the second network uses 8 bytes header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP headers are 20 bytes. (15) [CO3]

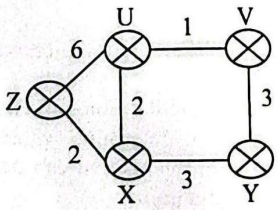
6. a) "IPv6 is faster and more secure than IPv4" – justify the statement based on IP datagram header. (05) [CO2]

   b) What are the differences between IP based forwarding and generalized forwarding? Explain simple packet handling rules for generalized forwarding based on OpenFlow protocol. (10) [CO3]

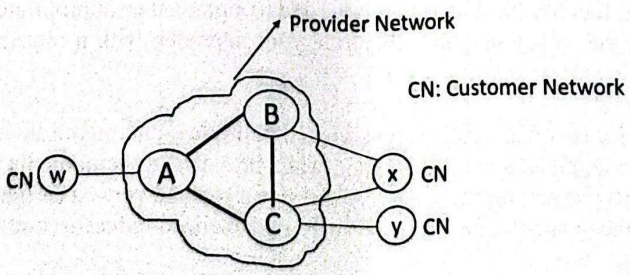
   c) What is link-local address in IPv6? Describe the process of Router Solicitation Message showing IPv6 header and ICMPv6 header values to get an IP using Stateless Auto Configuration mode. Calculate the IP of PC-A during this process shown in following figure. (20) [CO4]




7. a) Consider the network shown in following figure and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12) [CO3]



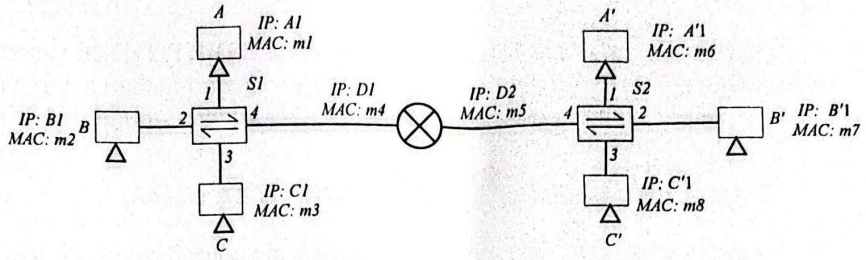

   b) What is Hot Potato routing? Describe the policy-based path selection for the following scenario. (10) [CO2]



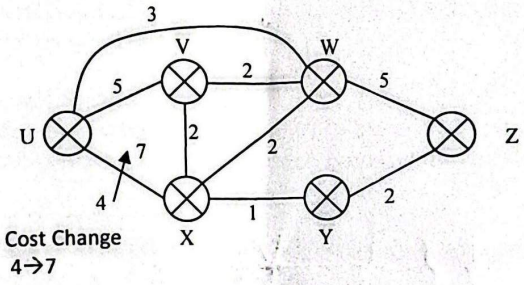



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c) You have to look carefully the following figure. Computer A wants to send a frame to Computer C'. Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? Initially switch table was empty. Finally, write down the self-learning algorithm of switch. (13) [CO2]




8. a) Explain Layer 2.5 protocol-based packet forwarding operation using an example. (10) [CO3]
   b) Emulate the Software Define Network (SDN) controller's interaction with local switches to modify the link-state changes occurred in the following network. Apply Dijkstra's link-state algorithm in the SDN controller to calculate the updated routing table for 'U'. (17) [CO4]




c) State how packet filtering firewall works. (08) [CO1]

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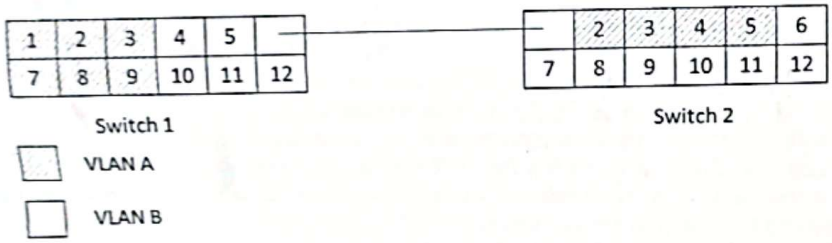
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## SECTION A
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the purpose of OSI model. Do all devices present in a network implement all OSI layers? Justify your answer. (10) [CO1]

   b) In the network shown in following figure, the server 223.1.2.5 receives an IP packet with src: 0.0.0.0, 68 and dest: 255.255.255.255, 67. What would happen next? Complete the interaction. (12) [CO2]

   c) Explain the benefits of Port-based VLANs. Explain the process and switch table to transmit a frame from port 2 of switch 1 to port 4 of switch 2, shown in following figure. (13) [CO3]

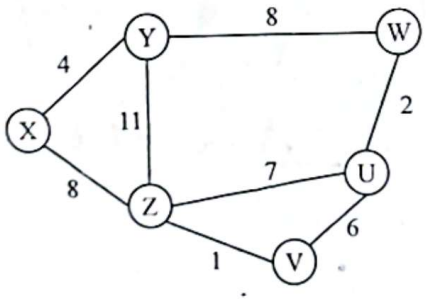



2. a) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10) [CO1]

| | | | |
|---|---|---|---|
| 4 | 5 | 0 | 28 |
| 1 | | 0 | 0 |
| 4 | 17 | 0 | |
| 10.12.14.5 | | | |
| 12.6.7.9 | | | |

   b) Explain the hot potato routing protocol. Briefly explain the benefits and drawbacks of this protocol. (12) [CO3]

   c) Using a Link-State routing algorithm, determine the routing table at the router X from the topology given in following figure. (13) [CO2]




3. a) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 group of customers: (13) [CO2]
   i) 1st group: 52 customers, each needs 128 addresses.
   ii) 2nd group: 40 customers, each needs 256 addresses.
   iii) 3rd group: 100 customers, each needs 64 addresses.

   How many addresses are still unallocated? Note down the network addressed of each group.

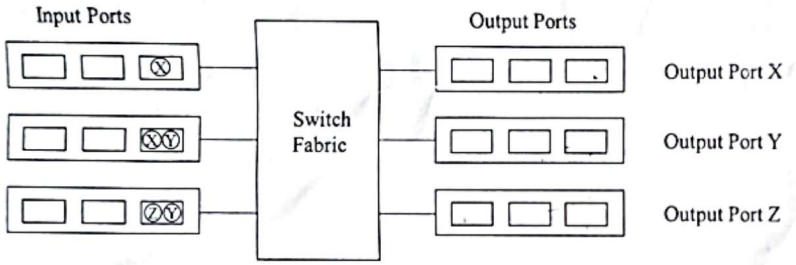


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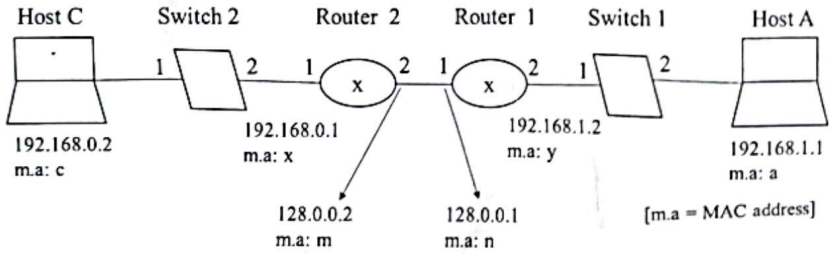
b) Suppose you have a broadcast channel in Ethernet CSMA/CD algorithm with transmission rate 10 Mbps where a node can transmit a frame of max 1500 bytes, and maximum propagation delay between 2 nodes is: 2ms. (10) [CO]
   i) Calculate the efficiency of the algorithm.
   ii) When would it reach its highest efficiency, and why?

c) In the switch shown in following figure, all datagrams have the same fixed length and the switch operates in a slotted, synchronous manner. Assuming a datagram can be transferred from input port to output port in one time slot, what is the minimal number of slots needed for the transfer of all datagrams when: (12) [CO1]
   i) The switching fabric is memory.
   ii) The switching fabric is bus.
   iii) The switching fabric is crossbar.

   Briefly explain the process for each of these switching fabrics (You can assume any input queue scheduling order).




4. a) Define SDN control plane. Briefly explain the characteristics of SDN. (10) [CO1]
   b) State router oscillation problem with proper example. When it occurs? (10) [CO3]
   c) In the network shown in following figure, a data packet is routing from A to C and C sends a response to A. Assuming all hosts know the necessary IP addresses, MAC addresses and first hop router addresses, show the data capsulation and decapsulation in each device with necessary diagram(s) and table(s). (15) [CO2]




## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) List down the steps involved in SSL handshaking. Also, present a pictorial view of packet transmission. (10) [CO2]
   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10) [CO2]
   c) Suppose a sender and receiver are using a pipelined protocol, i.e., Go Back-N or Selective Repeat. In this specific case, the rate at which the sender process pushes data is less than the rate at which then ~~sender~~ receiver consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. (05) [CO3]
   d) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Explain the procedure of getting IP address using recursive queries for various DNS servers. (10) [CO2]

6. a) State the advantage of P2P over client-server architecture. Despite those disadvantages, why do we find Client-server architecture applied in most of the real-life cases? (10) [CO1]
   b) Consider the following figure where Mr. X wants to access a webpage hosted on a Google server. This webpage comprises a base HTML file along with 10 JPEG images, each with a length of 1024 bits. The transmission rate of the link connecting Mr. X to the server is 1 Mbps. Determine the total response time for both Non-persistent and persistent HTTP by real internet delay. Mr. X measures real internet delay by traceroute program. (15) [CO2]



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```
C:\Users\HP>tracert google.com
Tracing route to google.com [142.250.193.78]
over a maximum of 30 hops:
  1    <1 ms    <1 ms    <1 ms  10.24.241.1
                  .
                  .
                  .
 14    82 ms    80 ms    80 ms  142.251.54.83
 15    80 ms    79 ms    78 ms  del11s16-in-f14.1e100.net
Trace complete
```

c) Consider a reliable data transfer protocol that uses only negative acknowledgements. Suppose the sender sends data only infrequently. Would a NAK-only protocol be preferable to a protocol that uses ACK? Why? Now suppose the sender has a lot of data to send and the end-to-end connection experiences few losses. In this second case, would a NAK only protocol be preferable to a protocol that uses ACKs? Why? Justify your answer. (10) [CO3]

7. a) Suppose, a HTTP client is sending 1 bytes of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasoning. (15) [CO3]

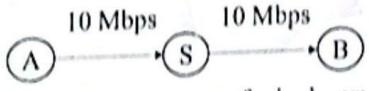
b) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (11) [CO3]

c) State how packet filtering firewall works. (09) [CO1]

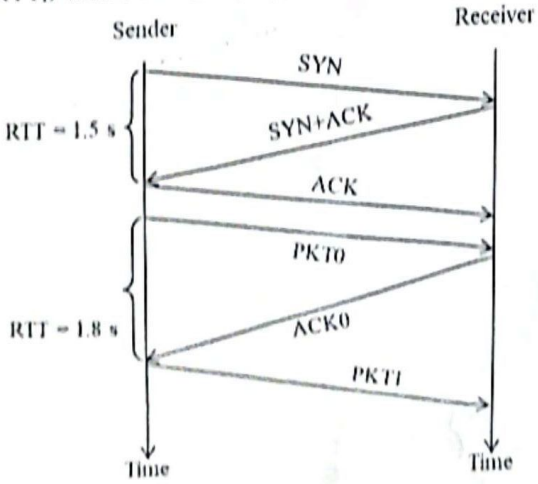
8. a) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html* and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (10) [CO2]

b) Host A and B are each connected to a switch S via 10 Mbps link as shown in following figure. (10) [CO1]

The propagation delay on each link is 20μs. Assume signaling message for circuit setup and acceptance are 100 bit long and processing time of a signaling message at switch S, hosts A and B is 15μs. These signaling messages are transmitted over the 10 Mbps link. Calculate the total time required to transmit a 10,000-bit message from A to B for circuit switching.




c) In the following figure, some desired packet transfer is shown. Calculate Retransmission Timeout time (RTO) where applicable. Let, initial RTO=6s. Recall that, $RTO = RTT_s + 4\,RTT_D$ where $RTT_D = (1-\beta)\,RTT_D + \beta\,|RTT_s - RTT_M|$ (15) [CO3]






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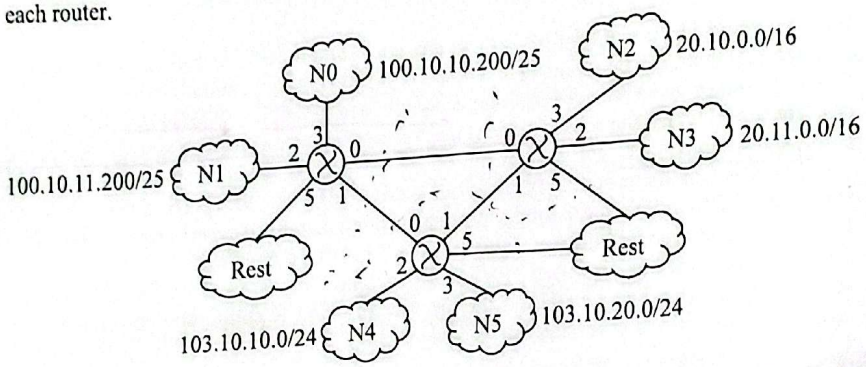
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## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define protocol with proper example. Do you think internet standards are necessary for defining computer network protocols? Why or why not? (10)
b) Let you are a network engineer at XYZ Inc. Your task is to design the forwarding tables of each router in a network. Consider the following network and design the forwarding tables of each router. (15)




c) Are there any relations between queuing and switching fabric speed in case of a router? Explain it. (10)

2. a) Define router oscillation problem with proper example. When it occurs? (11)
b) You are given an IPv4 datagram described in hexadecimal format: "4640 001C 00D1 00B9 1E11 DE0B 4AFA 0A14 C878 321E 0000 000F 0272 6F9A". Answer the following questions based on the given datagram (in decimal). (15)
i) Header length (in bytes) = ?
ii) Identifier = ?
iii) Fragment offset = ?
iv) Payload = ?
v) Is it a fragmented packet? If yes, does the packet has any specialty?
vi) Which fields will get updated if the packet is sent to the next hop router?
c) How does classless addressing solve the problem with classful addressing? (09)

3. a) Describe the services offered by network layer. Discuss the necessities of IPv6 over Ipv4. (10)
b) Why does collision occur in CSMA when we only send the packets if the channel sensed idle? Also, demonstrate the collision detection mechanism and algorithm to resolve the issue. (16)
c) "In BGP, packets are not routed to a specific destination address, but instead to CIDRized prefixes"—justify the statement. (09)

4. a) State how packet filtering firewall works. (06)
b) Demonstrate how NAT operates. What are the controversial things behind NAT? (09)
c) Explain how TLS functions. Mention at least one method of obtaining the four keys from the master key. (08)
d) Let you want to send an email to your friend securely. By ensuring the three most important properties of secure communication, demonstrate the scenario for safeguarding email. (12)



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## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the advantages of Peer-to-Peer architecture over Client-Server architecture. Despite those advantages, why do we find Client-Server architecture applied in most of the real life cases? (10)

   b) Should your ISP have its own cache server? Justify your answer. Explain how a cache server acts as both a client and a server. (10)

   c) Say, there is a HTTP client sending 1 byte of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasonings. (15)

6. a) Does rdt 3.0 use same procedure in case of lost packet and lost acknowledgement? Explain with proper illustrations. (10)

   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10)

   c) Suppose, a client wants to start an online service. He has five computers ready for this job. These are connected to the internet with IP addresses 212.212.212.101–105. He picked "Network utopia.com" as the domain name for the online service. Your job is to make the service accessible by anyone on the internet. What will be your steps to accomplish that? Explain briefly. (15)

7. a) What are port addresses used for? Is there any constraint on how to assign port addresses or how many of them can be assigned at a time? If yes, then discuss about them. (10)

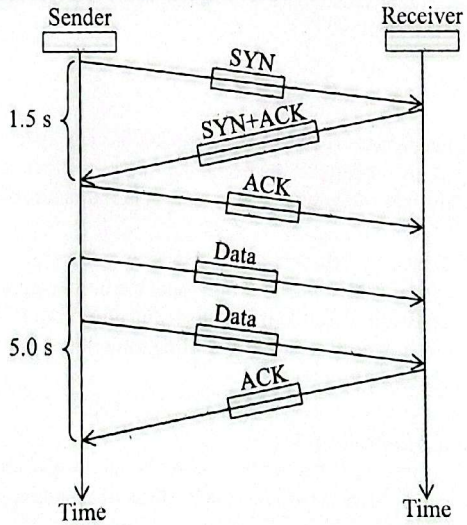
   b) Suppose a sender and a receiver are using a pipelined protocol, i.e., GBN or SR. In this specific case, the rate at which the sender process pushes data is less than the rate at which the receiver process consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. Also, say, the speed of transport layer at sender side is greater than the sender process pushing data to transport layer. In that case, is there a need for a sender buffer at transport layer? Justify your answer. (13)

   c) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html*, and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (12)

8. a) Differentiate between flow control and congestion control. Explain the necessity of either of those in a reliable data transfer protocol. (10)

   b) In the following figure, some desired packet transfer is shown. Introduce a retransmission timer and then calculate and show Retransmission Timeout time (RTO) where applicable. Let, the initial RTO = 6 s. (12)

   Recall that, $RTO = RTT_s + 4RTT_D$, where $RTT_D = (1-\beta)\, RTT_D + \beta \,|\, RTT_s - RTT_M|$. Let, the value of $\alpha$ to calculate $RTT_s$ is 0.125 and $\beta$ is 0.25.




   c) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (13)



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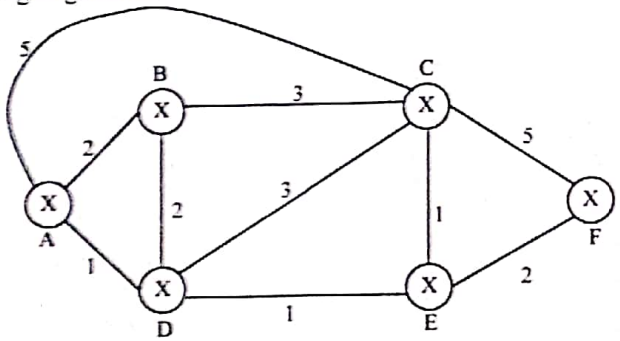
## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly discuss about client-server architecture. Discuss about the major challenges of P2P architecture. (12)
   b) "Socket is analogous to door" – Justify the statement. What transport service does an app need? (08)
   c) What are the components of cookies? Describe how cookies keep state using database with proper examples. (10)
   d) FTP control connection is "out of band". Explain. (05)
2. a) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Show, how it is done using recursive queries for various DNS servers. (09)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (10)
   c) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10)

| | | | |
|---|---|---|---|
| 4 | 5 | 0 | 28 |
| 1 | | 0 | 0 |
| 4 | 17 | 0 | |
| 10.12.14.5 | | | |
| 12.6.7.9 | | | |

   d) How does conditional GET command improve performance of web proxy and local cache mechanism? (06)
3. a) Explain how DHCP works. What are the main benefits of DHCP? (09)
   b) Apply fragmentation on following information for IPv4 datagram. (13)
      i) Original Datagram: 5000 bytes, Header: 20 bytes.
      ii) MTU: 1st level 2000 bytes & 2nd level: 800 bytes.
   c) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 groups of customers: (13)
      i) 1st group: 52 customers, each needs 128 addresses.
      ii) 2nd group: 40 customers, each needs 256 addresses.
      iii) 3rd group: 100 customers, each needs 64 addresses. How many addresses are still unallocated? Note down the network addresses of each group.
4. a) Briefly explain how PAT works. Why is PAT used over NAT? (10)
   b) Determine the routing table/forwarding table at the router B from the following topology using a Link-state Routing Algorithm. (17)




   c) Concisely describe the error reporting messages of ICMP. (08)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (10)
   b) Write filtering/forwarding algorithm of a switch. List the actions with corresponding conditions that a switch may perform when it receives a frame. (09)
   c) What is multiprotocol label switching (MPLS)? How does it work? Describe MPLS functionality with a simple example. (12)
   d) Describe silly windows syndrome created by the sender. (04)

6. a) How does TCP handle congestion? Describe with proper state diagram and graph. (15)
   b) Write down the connection establishment and half close in TCP. (10)
   c) How retransmission timeout is calculated in TCP? In which situation Karn's algorithm is applied? (10)

7. a) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (10)
   b) What is the difference between RDT 2.2 and RDT 3.0? Describe the FSM of RDT 3.0. (12)
   c) Write short notes on (i) Persistent Timer and (ii) Keepalive Timer. (08)
   d) What should be the relation between window sizes and sequence number in case of Selective Repeat? Why do we need the relationship? (05)

8. a) What are the shortcomings of symmetric key cryptography? How are those solved in public key cryptography. (06)
   b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)
   c) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (07)
   d) Suppose, A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (12)
      i) A and B are in the same network.
      ii) A and B are in the different network.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) What does protocol achieve? Differentiate between intranet and extranet. (10)
   b) Why do we need Web caching? Discuss about the consistency of Web caching. (10)
   c) What is P2P architecture. Describe what transport service does an app need? (10)

2. a) Compare among WWW, HTTP and Internet. Show the HTTP request and response message format. (10)
   b) Suppose, *Alice* (ID – "alice@net.edu") wants to send message to *Bob* (ID – "bob@denza.edu"). Show SMTP interactions between SMTP client and SMTP server for this communication. (20)

3. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (14)
   b) How does RSA provide digital signature mechanism? (10)
   c) Justify that – "Switch is more intelligent than hub". (06)

## SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Why do we need Network Address Translation (NAT)? Briefly explain how does NAT work. (10)
   b) Compare the contrast link-state and distance vector routing algorithms. (09)
   c) Suppose a source and a destination are connected through a switch and a router. If the source sends a message to the destination, then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above-mentioned devices. Explain with necessary figure. (11)

5. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) What is the relation between window size and sequence number in case of selective repeat? Why this relation is needed? (09)
   c) List down the steps involved in SSL handshaking. Also present a pictorial view of packet transmission. (10)

6. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) What is Ethernet? Describe how does CSMA/CD protocol work with a detail look on exponential back-off algorithm. (12)
   c) Suppose, a company has 200 personal computers and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PCs. How personal computer get the IP address if you have a network address: 172.16.0.0/24. Also, state the dynamic range of IP addresses. (12)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

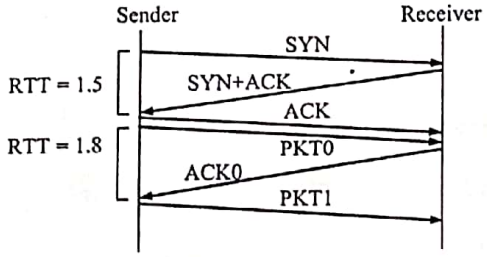
## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is access network? Describe the enterprize access network with diagram. (08)
   b) How does data pass through network in case of circuit switching? Explain in brief. (08)
   c) What is troughput? What will be the troughput in case of Internet scenario? (06)
   d) What is the differences between RDT 2.1 and RDT 2.2? Draw RDT 2.2 finite state machine for sender and receiver. Briefly describe this. (13)

2. a) Why is UDP used in transport layer though it is unreliable? (05)
   b) In GB3, if every 5th packet is being transmitted is lost and if we have to send 10 packets, then how many transmissions are required? (11)
   c) What is the relation between window size and sequence number in case of selective repeat? Why is this relation needed? (10)
   d) Write short notes on the following terms: (09)
      (i) Demultiplexing, (ii) Characteristics of UDP, and (iii) UDP segment format.

3. a) What is "syn-flooding" attack in case of connection establishment in TCP? Explain. (06)
   b) How many ways are there to terminate a connection in TCP? Explain the ways in proper diagram. (12)
   c) Suppose receiver application program consumes data as 1byte at a time. For this, the receiver TCP announces the window size as 1byte. So, the sending TCP sends 1byte at a time and the traffic intensity will be high. What has to be done to solve the problem? Explain two solutions. (06)
   d) Explain Karn's algorithm in case of TCP timer. What will be the RTO of PKT1 in the following figure? (11)




4. a) How does TCP recognize the network congestion? Explain the Reno approach of managing this congestion by using necessary illustration. (13)
   b) Explain block cipher and cipher-block chain algorithm in symmetric key cryptography. Why will we use cipher-block chain instead of block cipher? (14)
   c) In case of RSA, if primes are $p = 5, q = 7$, public key is (35,5) and cipher text $c = 18$, then decrypt the cipher text to get plain text. (08)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose two nodes start to transmit at the same time a packet of length $L$ over a broadcast channel of rate $R$. Denote the propagation delay between the two nodes as $d_{prop}$. What there be a collision if $d_{prop} > \frac{L}{R}$? Why or why not. (08)
   b) Briefly describe that two dimensional parity checks can correct and detect a single bit error but a double bit error that can be detected but not corrected. (08)

c) Why is an ARP query sent within a broadcast frame? Why is an ARP repose sent within a frame with a specific MAC address? (07)

d) You have to look carefully Figure 5(d). Computer A wants to send a frame to computer $C'$. Describe the two switches' behavior when A connects to $C'$ and after completing the communication, how the switch table of both switches look like? (Initially switch table was empty). Finally, write down the self-learning algorithm of switch. (12)

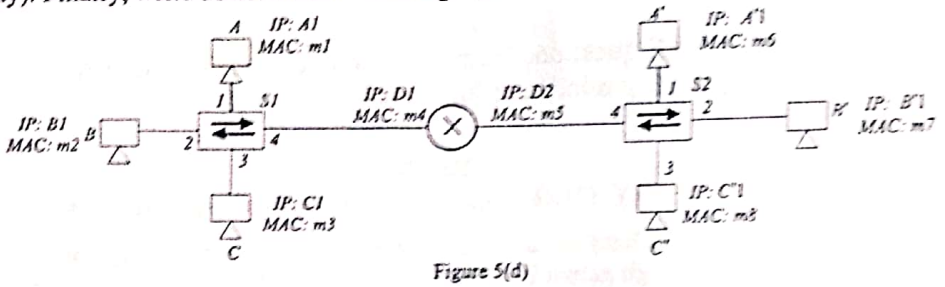



Figure 5(d)

6. a) Suppose you wanted to do a transaction securely from a remote client to a server as fast as possible. Would you use UDP or TCP? Why? (07)

b) What is meant by handshaking protocol? Why is it said that FTP sends control information "out of band"? (07)

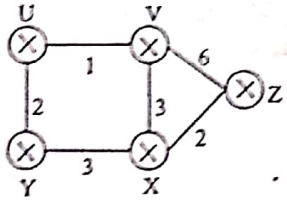
c) What are the advantages and disadvantages of a circular DHT (with no shortcuts)? (08)

d) Suppose the CSE department of KUET has 200 personal computer and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PC. How personal computer get the IP address if you have a network address 172.16.0.0/24. Also, state the dynamic range of IP addresses. (13)

7. a) Briefly describe different types of switching fabrics. Which, if any, can send multiple packets across the fabric in parallel? (06)

b) Define HOL blocking and longest prefix matching with appropriate example. (08)

c) Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12)




d) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (09)

8. a) What is NAT traversal problem? How can you solve it? Explain briefly with appropriate figure. (11)

b) Compare and contrast link-state and distance vector routing algorithms. (08)

c) Describe the technologies that are used for transitioning from IPv6 to IPv4 to IPv6. (07)

d) Consider the generator G = 10011 and suppose that D has the value 1010101010. What is the value of R? How can you detect the error using Cyclic Redundancy Check method? (09)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) State the problems of a stop-n-wait type reliable data transfer protocol. Design a reliable, pipelined data transfer protocol that uses no buffer for out of order packets. (15)
   b) Briefly describe the services provided by TCP. (09)
   c) Show the format of HTTP request line. Mention at least two request type used by HTTP version 1.1 and explain. (06)
   d) Explain common FTP commands. (05)

2. a) Consider transferring an enormous file of $L$ bytes from host A to host B. What is the maximum value of $L$ such that TCP sequence numbers are not exhausted? Assume an MSS of 1460 bytes. (05)
   b) The TCP of a host receives five segments at the time shown (HH:MM:SS.mmm) below. If it has no data to send, show the acknowledgements sent by the host using line diagram. (09)
   (i) at 0:0:0.000 (ii) at 0:0:0.027 (iii) at 0:0:0.400 (iv) at 0:0:1.200 (v) at 0:0:1.208
   c) Show a sample SMTP interaction between SMTP client and server to transfer the message "Good morning my friend" from alice@xxx.com to bob@yyy.com. (11)
   d) In TCP, what problems are solved by the following timers and how? (10)
   (i) Persistence timer (ii) Time-waited timer

3. a) Suppose a DNS client is looking for the IP address of www.kuet.ac.bd. It already sent a query message to the DNS server. Show the response message sent by the server. Use necessary DNS compression. Assume the IP address is 128.175.13.63. (10)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (09)
   c) Since UDP is unreliable, why is it used at transport layer? (05)
   d) What do you mean by digital signature? How does RSA provides digital signature mechanism? (11)

4. a) (15)

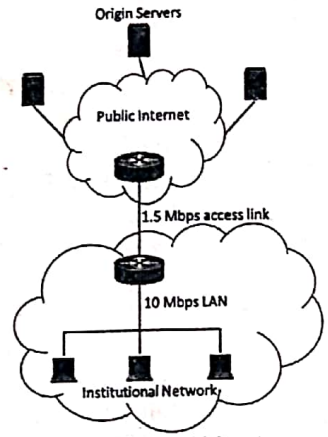



Figure 4(a)

Consider Figure 4(a), for which there is an institutional network connected to the internet. Let the average object size is 900 kbit and the average request rate from institutional browsers to the origin servers is 1.5 request per second. Also suppose that the amount of time it takes from when the router on the internet side of the access link forwards an HTTP request until it receives the response in two seconds on average. Model the total average response time as a sum of the average access delay and the average internet delay. For average access delay, use $\Delta/(1-\Delta\beta)$, where $\Delta$ is the time required to send an object over the access link and $\beta$ is the arrival rate of objects to the access link.

i) Find the total average response time.
ii) Now suppose a web cache is installed in the institutional LAN. Let the hit rate is 0.4. Find the total response time.

b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)

c) List down the steps involved in SSL handshaking. Also present a pictorial view of packets transmission. (10)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) How does header checksum can be calculated in IPv4 datagram? Why does it cover only data portion? Explain with example. (12)

b) Define Computer Network. What are the major functionalities of Computer Network? (08)

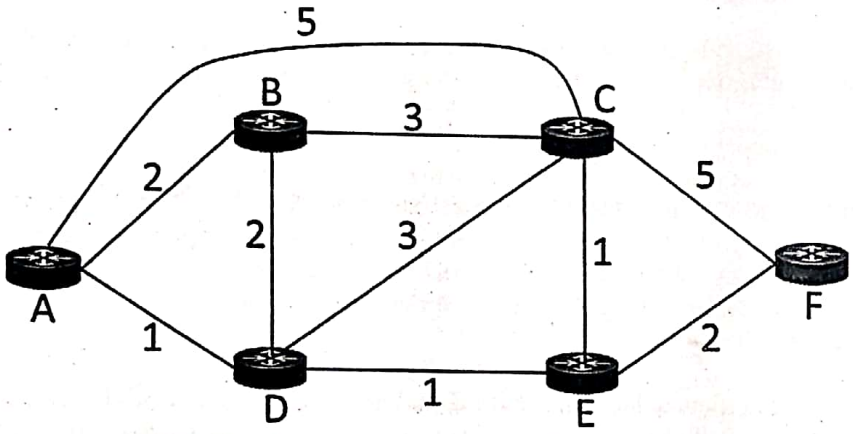
c) Apply Fragmentation on following information for IPv4 Datagram. (15)

i) Original Datagram : 5000 bytes, Header : 20 bytes.

ii) MTU : First level 2000 bytes and Second level 800 bytes.

6. a) Concisely describe the error reporting messages of ICMP. (10)

b) Determine the routing table/forwarding table at the router B from the following topology using a Link-State Routing Algorithm. (17)




c) "Switch is more intelligent than hub" – justify the statement. (08)

7. a) A router inside an organization receives a packet with the destination 190.240.33.91. How does it find the subnetwork address to route the packet? Assume the subnet mask is /19. (05)

b) Write a short note on Address Resolution Protocol (ARP). (09)

c) What is count to infinity problem? Use a suitable example to show that the poisoned reverse technique can solve the problem. (12)

d) Why do we need Network Address Translation (NAT)? Briefly explain how NAT works. (09)

8. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit $R$. (12)

b) How does Slotted ALOHA random access MAC protocol work? Show the efficiency of it. (13)

c) Write down the services provided by the data link layer. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2017
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

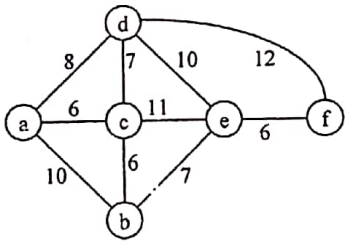
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define statistical multiplexing. Which one is better between statistical multiplexing and Time Division Multiplexing? Explain with an appropriate example. (12)
   b) Suppose source and destination are connected through a switch and a router. If source sends a message to destination then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above mentioned devices? Explain with necessary figure. (11)
   c) Suppose A is transmitting a packet to B. They are directly connected through a physical link. The length of the link is 1 km, the length of the packet is L bits and the transmission rate is 1 Mbps. (i) Now calculate how many bits will be arrived at B in 50 μsec? (ii) If the packet length is 100 bytes then at what transmission rate R, the propagation delay would be equal to the transmission delay of packet? (12)

2. a) Consider sending a 2400 byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? (10)
   b) Suppose the CSE department of KUET has 200 personal computers. The network administrator does not want to set the IP addresses statically. So, how the personal computers can get IP addresses? Explain steps of getting IP. (11)
   c) Consider the following network. With the indicated link costs, use Dijkstra's shortest path algorithm to compute the shortest path from *b* to all network nodes. Show how the algorithm works by computing a table. (14)




3. a) How does private IP addressing concepts resolve IPv4 address space problem? What are the new problems arises? Explain. (09)
   b) Suppose A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (13)
      i) A and B are in the same network.
      ii) A and B are in the different network.
   c) What is Ethernet? Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (13)

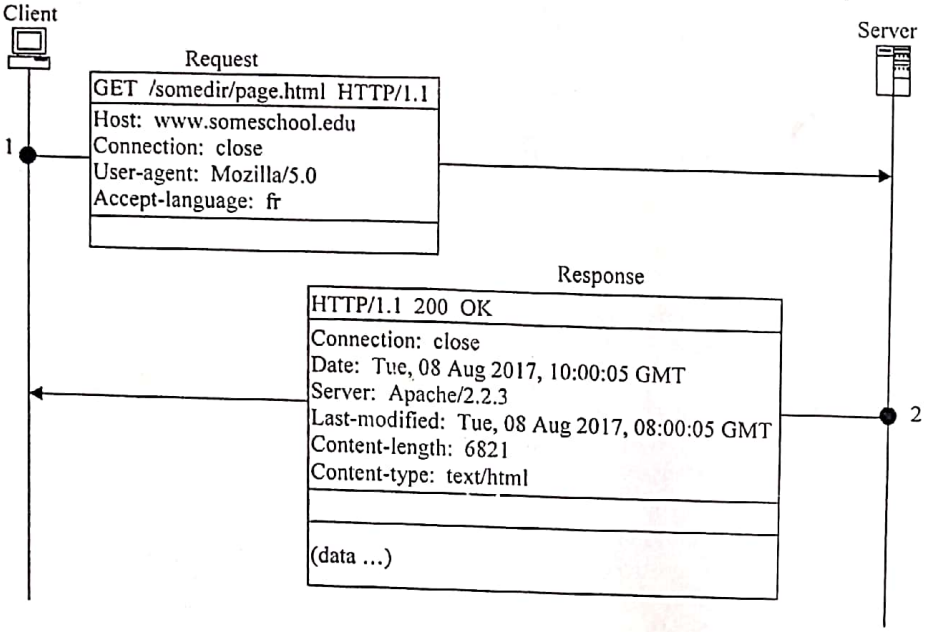
4. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (08)
   c) How does IEEE 802.11 deal with data fragmentation? Discuss. (10)
   d) The CSE department of KUET just bought a Cisco switch which has 50 ports. The network administrator connected 40 personal computers to that switch. Now explain the learning mechanism of it. (11)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

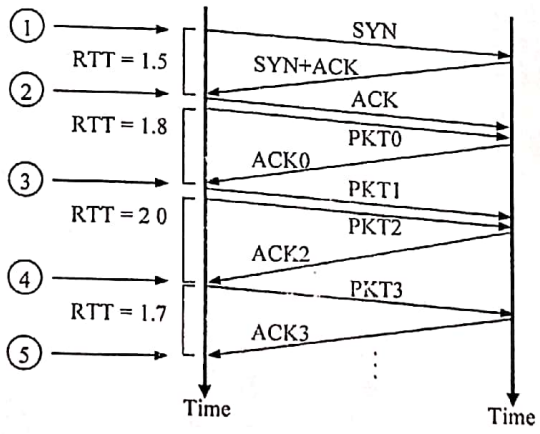
5. a) Compare among the Internet, WWW and HTTP. (09)
   b) What are the limitations of SMTP? How can these be solved by Multipurpose Internet Mail Extensions (MIME)? Briefly explain the working process and *Base64* encoding of MIME. (11)

c) Consider the following communication between a client and a server. The figure shows the different messages transferred between them. Explain each section as well as the statements that are presented in the messages. Also provide a general format of those messages. (15)




6. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) Design a stop-n-wait type reliable data transfer protocol by showing a finite state machine. Also present the line diagrams to explain its operation for various packet transmission scenarios. (14)
   c) How does selective repeat protocol work? Explain. (10)

7. a) How does TCP recognize the network congestion? Explain, with necessary illustration, the Reno approach of managing this congestion. (14)
   b) What is silly window syndrome? How does TCP handle this issue? (08)
   c) How does TCP set the timeout interval for retransmission timer? What are the Jacobson/Karels proposals for setting that interval? Consider the following communication scenario of TCP. Figure shows the sample RTTs in seconds. Compute the timeout intervals at the points shown by arrows i.e. at initial position and at the end of each acknowledge received. (13)




8. a) Design a protocol to authenticate a live entity. Can it be used for public key system? Why or why not? (12)
   b) Let you are a network administrator, design a rule set for your packet filter firewall to allow your local users only to browse outside and no other traffic except the locally initiated return traffic from outside is allowed to enter your network. (07)
   c) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (09)
   d) What are the basic differences between TCP and UDP? (07)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2016
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CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

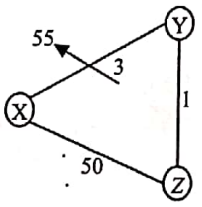
(Answer **ANY THREE** questions from this section in Script A)

1. a) What is layering in computer networks? Write down the major functions of network layer. (07)
   b) What are the advantages of packet switching over circuit switching? (06)
   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14-bytes headers and has an MTU of 1024 bytes; the second user 8-byte header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP header are 20 bytes. (12)
   d) Differentiate between IPv4 and IPv6 header format. Explain the technologies that are used for transition from IPv4 to IPv6. (10)

2. a) Write down the pros and cons of static and dynamic routing. Explain link-state routing algorithm with a proper example. (13)
   b) What is DHCP? How does DHCP work? Write down the advantages and disadvantages of DHCP. (12)
   c) What does ICMP stand for? What does it do? Describe the error reporting messages of ICMP. (10)

3. a) "A transport-layer protocol provides logical communication between processes running on different hosts, a network-layer protocol provides logical communication between hosts."- Justify the statement. (08)
   b) What are the essential tools of RDT protocol? Describe the working mechanism of RDT 2.0. How can the performance problem of RDT be solved? (13)
   c) Describe the TCP timers in Brief. (08)
   d) Since UDP is unreliable, why is it used at transport layer? (06)

4. a) Consider the following graph. Assume that link $(x,y)$ cost has been changed from 3 to 55. Show that the poissoned reverse technique can resolve the count to infinity problem in this case. (11)




   b) In theory TCP is fair, but in face it is not. Give an example. (07)
   c) What is the NAT traversal problem for P2P applications, say Skype? (05)
   d) Suppose the values of $p$ and $q$ are 7 and 13 respectively. Find suitable public and private keys for RSA algorithm. Verify the keys for the message "CSE". (12)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the techniques to resolve IPv4 address space problem? Explain in Brief. (10)
   b) What is digital signature? How authentication is provided using digital signature. (10)
   c) What is Diffie Hellman key exchange algorithm? Explain its working principle with an example. (10)
   d) What do firewalls do? Explain shortly. (05)

6. a) "Sockets are the software interface"-Justify the statement. (05)
   b) What are the transport services does an application need? Briefly describe them. (10)
   c) What is the benefit of stateless property of HTTP server? How HTTP servers identify users with stateless property? Explain briefly. (10)
   d) Differentiate GET and Conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cash mechanism? (10)

7. a) Why is it said that FTP sends control information "out of band"? (05)
   b) Suppose Alice, with a web-based email account (such as gmail) sends a message to Bob, who accesses his mail from his mail server using POP3. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application layer protocols that are used to move the message between the two hosts. (10)
   c) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (08)
   d) What is service provided by DNS system? Explain load distribution briefly. (06)
   e) If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Why or why not? (06)

8. a) Consider an e-commerce site that wants to keep a purchase record for each of its customer. Describe how this can be done with cookies? (08)
   b) Why are content distribution networks (CDNs) important in internet system? Describe the role of DNS in CDN. (07)
   c) Write down the advantages and disadvantages of polling and Token passing MAC protocol. (09)
   d) Discuss Byte stuffing with an appropriate example. (06)
   e) Why a network interface card is masked as semi-autonomous device? (05)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2015
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

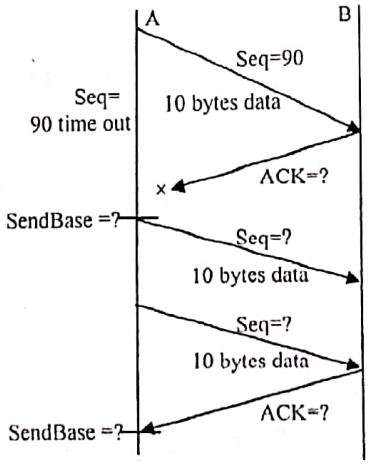
1. a) Define computer network and Internet. Explain Internet on service view points. (09)
   b) Why is it said that packet switching employs statistical multiplexing? Contrast statistical multiplexing with the multiplexing that takes place in TDM and FDM. (09)
   c) "Message segmentation reduces the end to end delay increasing resource usabilities"- Justify the statement. (08)
   d) Explain Internet structure with different tiers. What is the benefit of tierwise ISP connections for internet service? (09)

2. a) Why does TCP socket identification need 4-tuple instead of 2-tuple? What are those tuples? (08)
   b) Why UDP is called a bare-bone protocol in compare of TCP? (06)
   c) In respect of Rdt 2.0, briefly state the objective of the following used techniques: (i) Checksum (ii) ACK packet (iii) NAK packet. (09)
   d) Consider sending a file of 10KB packet with 1 Gbps bandwidth link and 15 msec end-to-end propagation delay. Answer the following questions: (12)
      (i) What is the required transmission time?
      (ii) What is the channel utilization (U)?

Note that $U = \dfrac{\frac{L}{R}}{RTT + \frac{L}{R}}$

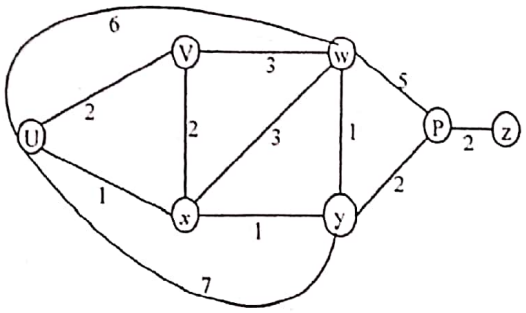
(iii) For getting the channel utilization 0.5 what will be the transmission time?

3. a) What is the performance problem of Rdt 3.0 (stop-and-wait) protocol? How this problem can be resolved? (07)
   b) Sketch a diagram where slective repeat protocol experiences a dilemma. Give your opinion how this dilemma can be overcome. (10)
   c) Consider TCP's communication scenario between A and B. A and B have initial sequence number 90 and 180 respectively. A is sending a file to B with a packet size of 10 bytes in the following figure. Determine the values of seq#, ACK# and sendBase#. (08)

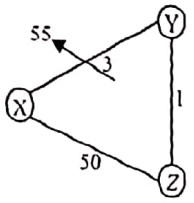



   d) What do you mean by NAT? Explain how NAT operation is performed. (10)

4. a) What is the HOL blocking in the switch fabric of a router? (05)
   b) Consider the following graph. Using Link-state algorithm find the shortest-path tree for router U. (12)




   c) Consider the following graph. Assume that link $(x,y)$ cost has been changed from 3 to 55. Show that the poisoned reverse technique can resolve the count to infinity problem in this case. (11)




   d) Is RIP Intra-AS or Inter-AS routing protocol? Show a simple example of RIP routing table entries. (07)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Briefly explain Nonpersistent HTTP and Persistent HTTP with/without pipeline. Compare TCP connections among the three HTTP methods for a HTML file containing 10 other objects' link. (10)
   b) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (08)
   c) Does web browsing require UDP service along with TCP? Explain briefly. (05)
   d) "Email demands both push and pull operations"-Justify the statement. (06)
   e) Compare SMTP, HTTP and DNS on the basis of direct transfer and transfer through intermediate hosts. (06)

6. a) Explain common FTP commands and replies. (07)
   b) Explain importance of DNS in service view points. Also explain iterative and recursive DNS queries with appropriate example. (10)
   c) How DNS and CDN together and alone provide better web browsing facilities? (08)
   d) Explain importance of Network Operations Communications Center (NOCC) in content Distribution Network, (05)
   e) "In computer networking switch is transparent to hosts but router is not"-is it true? Justify your answer. (05)

7. a) Why a network interface card is a marked as semi autonomous device? (05)
   b) Explain ideal Multiple Access (MAC) protocol scenario. Compare three different MAC protocols. (09)
   c) Distinguish slotted ALOHA and pure ALOHA protocols. (06)
   d) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (09)
   e) Compare routing switches and hubs on operational viewpoints. (06)

8. a) Why does a computer need an MAC address, inspite of having an IP address? (05)
   b) "MAC address of a host never goes outside the network"-is it true? Justify your answer. (05)
   c) Briefly explain various desirable properties of secure communication. (06)
   d) What are shortcomings of symmetric key cryptography? How are those solved in public key cryptography? (07)
   e) What do you mean by digital signature? How does RSA provides digital signature mechanism? (12)